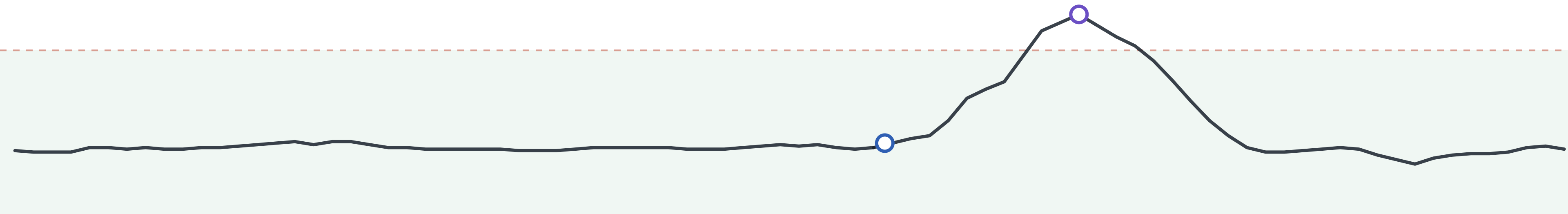


Glycemic Twin

A Digital Twin of **one person's** glucose response to meals, not just a prediction model.

State · prediction · explanation · what-if · history, built on real CGMacros v1.0.0 data



IdeaForge

TEAM LEADER

Mitansh Mathur

COLLEGE

IIIT Allahabad

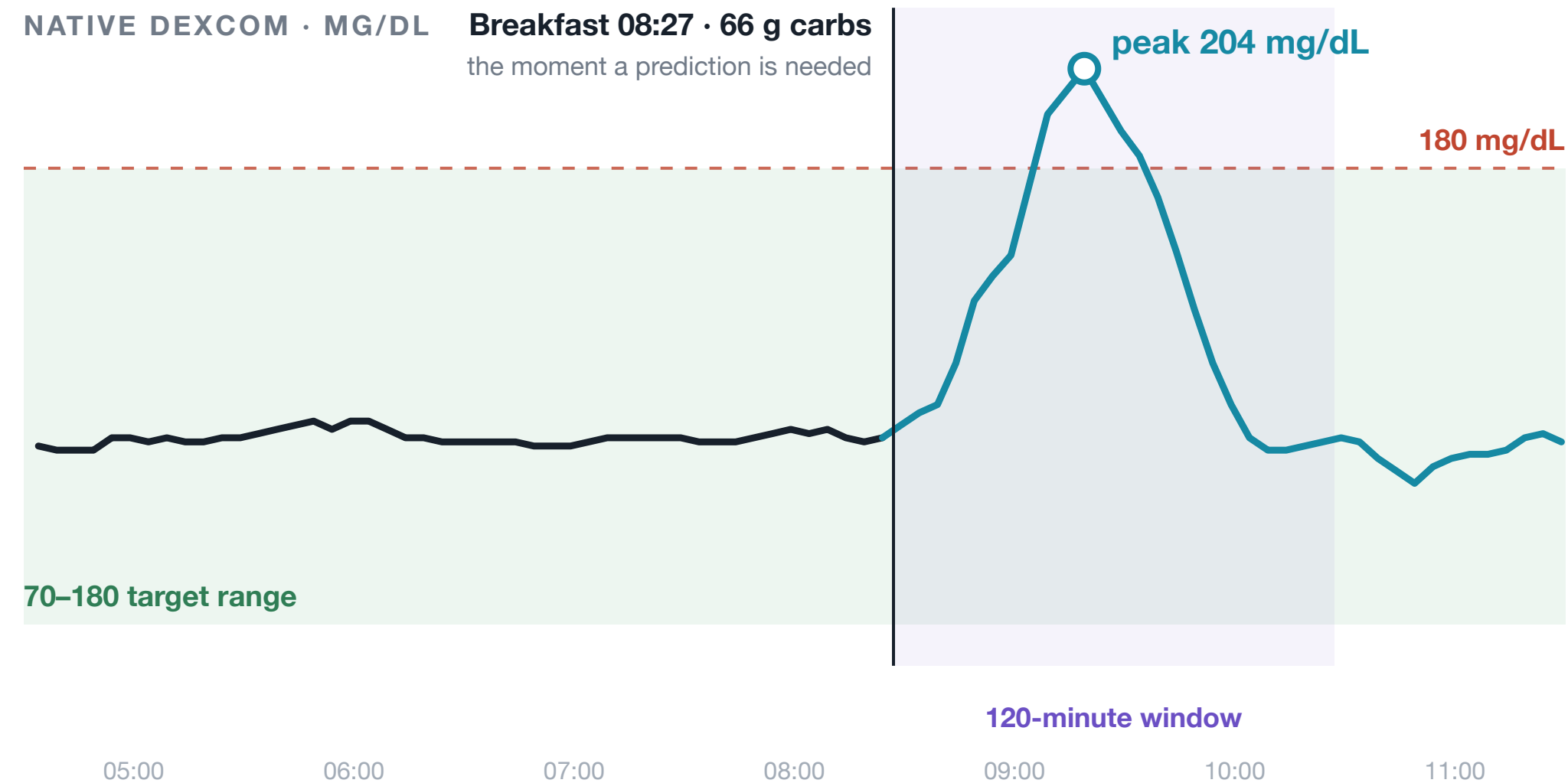
BUILT END TO END

Data audit → **Leakage-controlled features** → **Evaluated ML** → **Digital Twin engine** → **Secure API + database** →
Clinician interface → **Docker · CI**

— THE PROBLEM

A CGM shows what **already** happened.

In prediabetes and type 2 diabetes, glucose often rises above 180 mg/dL after meals. The response depends on the person, not only the meal.



BEFORE THE OUTCOME, A CLINICIAN STILL CANNOT ASK

“What was **likely** to happen?”

“**Why?**”

“What if the meal **changed?**”

Decision support for reviewing a patient's meals and glucose

WHO Clinicians Diabetes educators, endocrinologists, primary care, reviewing CGM, meal and wearable data before or during follow-up.	FOR WHOM Prediabetes · Type 2 Groups derived from baseline HbA1c. Healthy participants are evaluated separately.	WHAT IT GIVES A per-patient Twin Per-meal risk with uncertainty, its drivers, the learned personal response, prediction vs. reality, bounded what-if, and replay.
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DELIBERATELY OUT OF SCOPE

No medication, insulin, diagnosis or treatment functionality. No automated actions. Not a medical device, not clinically validated, not live monitoring.

Each existing view answers part of the question

CGM REPORT

Shows what happened

Curves and time in range, after the fact. No estimate before the outcome, no reasons, no scenarios.

Answers: “What happened?”

POPULATION RULES

Ignores the individual

“Fewer carbs”, fixed limits. The same advice for every response pattern.

Answers: “What works on average?”

A SINGLE ML SCORE

One number

No history, personalization, uncertainty or traceability, and often evaluated with leakage.

Answers: “How likely?”

GLYCEMIC TWIN

A per-person state over time

State at any moment (as_of)

Calibrated prediction

Explanation + provenance

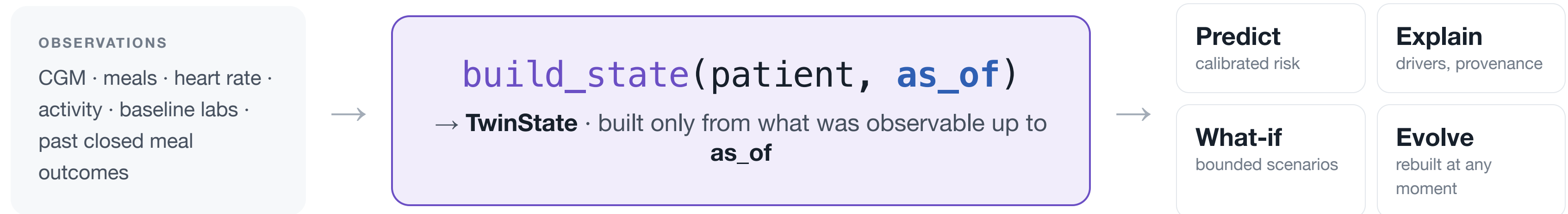
Bounded what-if

History: time rail, replay

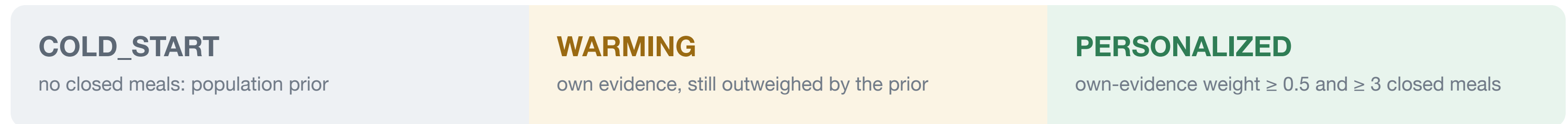
Answers: “How likely, why, what if, and how did it change?”

— THE CONCEPT

A Twin is a living per-person **state**, not a single score



LIFECYCLE · EVIDENCE ONLY ACCUMULATES AS TIME MOVES FORWARD



Real free-living data, about ten days per participant

44

participants with time series

1,663

logged meals with macronutrients

123,161

native Dexcom readings (every 5 min)

14 healthy · evaluated separately

16 prediabetes

14 type 2 diabetes

target group: 30 participants (groups derived from baseline HbA1c; no diagnosis field exists)

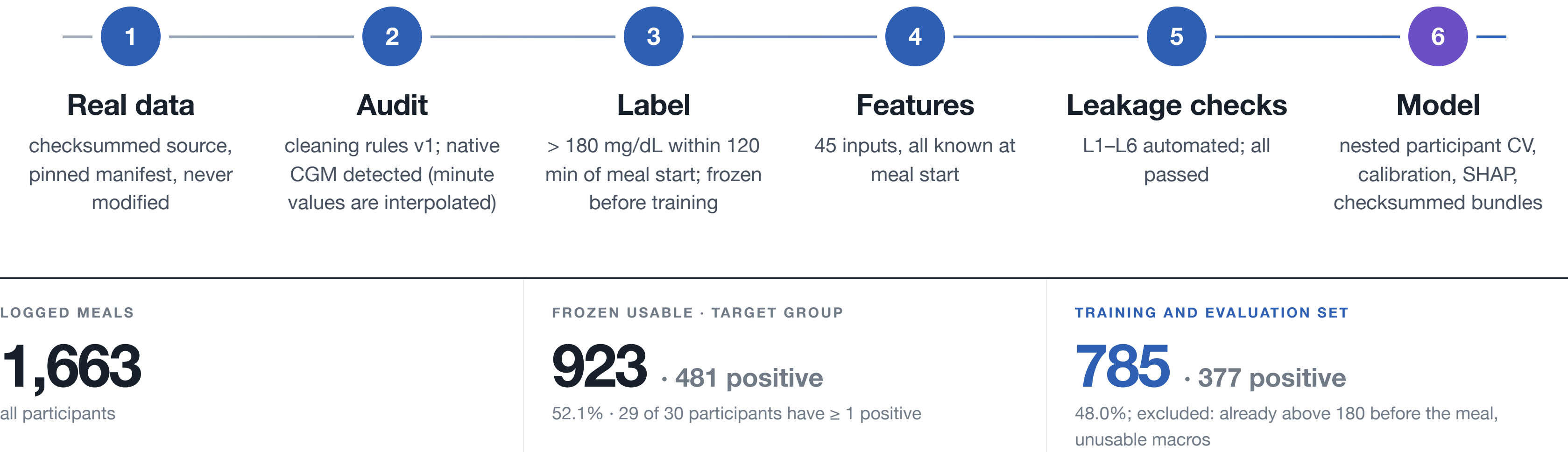
Dexcom G6 CGM

Fitbit heart rate + activity

Meal macros

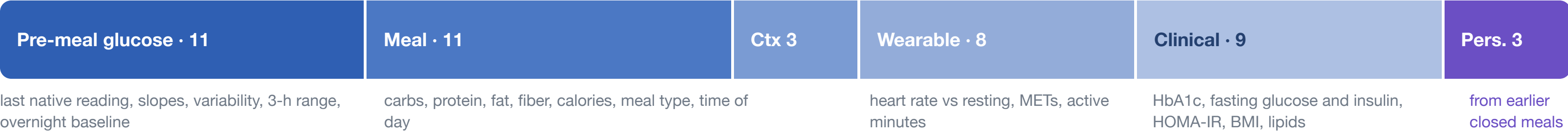
Baseline labs (HbA1c, fasting glucose, lipids)

From raw files to a model, with a report at every step

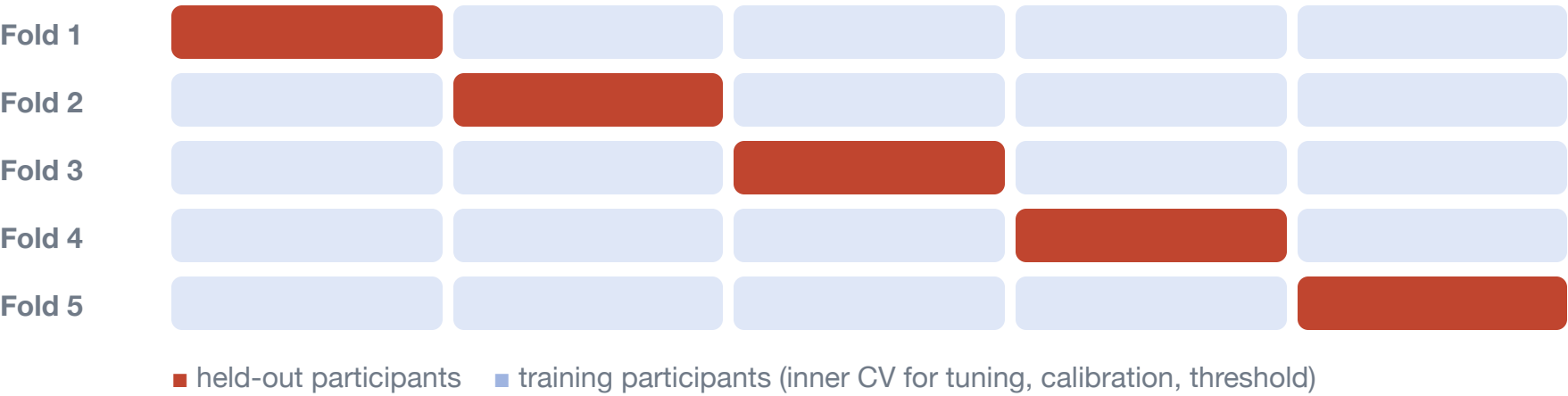


A frozen 45-input contract, evaluated on people the model never saw

FEATURE CONTRACT MODEL_FEATURES.V1 · 45 INPUTS, ALL KNOWN AT MEAL START



NESTED PARTICIPANT-LEVEL CROSS-VALIDATION · 5 PINNED FOLDS

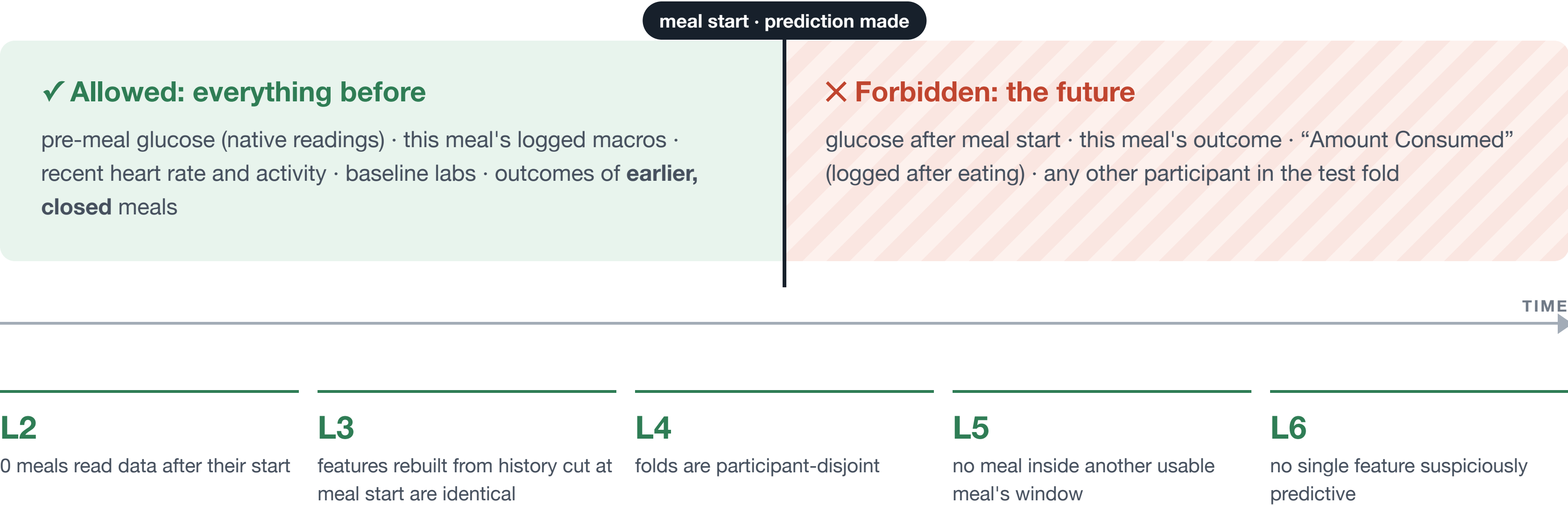


XGBoost and logistic regression, both calibrated; threshold chosen on inner folds.

Tuning, calibration, threshold and personal prior **never see the test participant**.

4 baselines, modality ablations, participant-bootstrap 95% CIs.

The model never sees the answer



PR-AUC · ACTIVE MODEL (XGBOOST, FULL + PERSONAL)

0.808

95% CI 0.736–0.870 · prevalence 0.480 (random model)

AUROC

0.804

0.759–0.845

CALIBRATION SLOPE

1.00

Brier 0.179

AT INNER-FOLD THRESHOLD

sens 0.806

spec 0.630

VS BEST SIMPLE BASELINE · GLUCOSE + CARBS RULE

0.701 +0.107

PR-AUC gain, paired participant bootstrap 95% CI 0.059–0.158 · AUROC +0.064

XGBOOST ≈ LOGISTIC REGRESSION

0.808 ≈ 0.807

difference +0.001 (CI –0.034 to 0.038). **Not claimed superior**; XGBoost is served, logistic is registered.

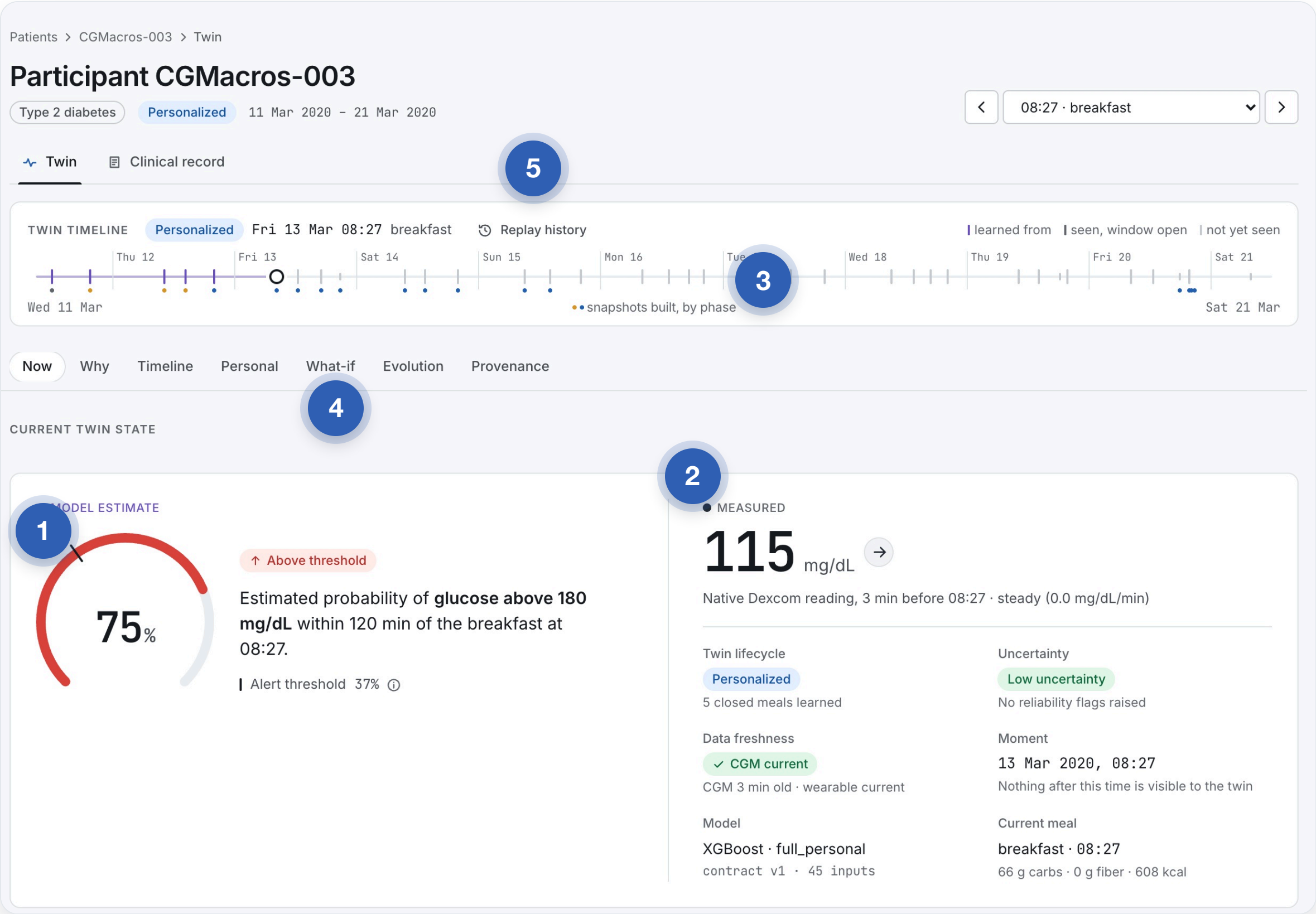
Combining data types helps: multimodal vs glucose-only +0.048 (0.010–0.091).
Personal prior +0.013 (–0.017–0.041): not proven.

One pure-Python Twin core, shared by training and serving



Provenance input and config hashes, model identity, engine version on every state	simulate() bounded, non-causal what-if on the current meal	diff_states() what changed between two moments, and why	Model hash bundle SHA-256 checked on every load; contract hash checked
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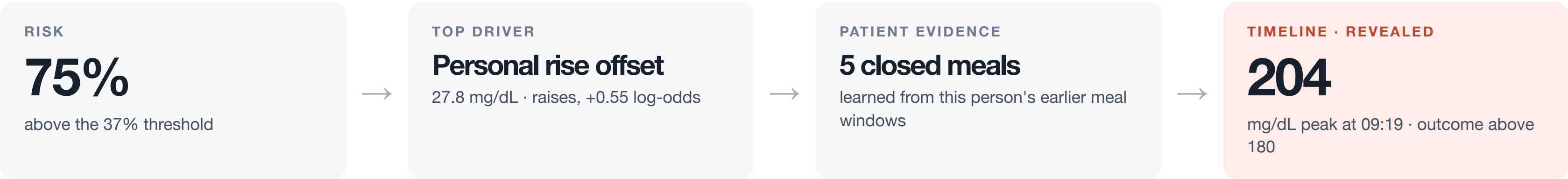
THE PRODUCT · PATIENT TWIN



See the Twin at a moment

- 1 Risk**
75% vs the 37% alert threshold, with uncertainty
- 2 Evidence**
measured glucose, lifecycle, freshness, model
- 3 Timeline**
learned from · window open · not yet seen
- 4 What-if**
bounded scenarios on this meal
- 5 Replay**
step the Twin through recorded days

Traceable, and checked against what happened



EXPLANATION

Why this risk?

Which inputs moved this estimate up or down, as computed by the served model for this exact prediction.

Model association · not causal

lowers the estimate

raises the estimate

bar length = share of the largest contribution

Personal rise offset

PERSONAL RESPONSE

27.8 mg/dL

How much more (or less) this person's glucose rose than the population model expects. Associated with a higher estimate for this meal.

Learned from this person's closed meal windows (Personal response). See personal response →

raises

+0.55

Protein

THIS MEAL

66 g

Logged protein of this meal. Associated with a lower estimate for this meal.

lowers

-0.54

Glucose variability, 1 h

GLUCOSE

1.1 mg/dL

Standard deviation of readings in the last hour. Associated with a higher estimate for this meal.

raises

+0.41

Carbohydrate

THIS MEAL

66 g

Logged carbohydrate of this meal. Associated with a higher estimate for this meal.

Taken from this meal's log.

raises

+0.30

Glucose at meal start

GLUCOSE

115 mg/dL

Last native Dexcom reading within 15 min before the meal. Associated with a lower estimate for this meal.

lowers

-0.24

Overnight baseline

GLUCOSE

114 mg/dL

Median glucose 02:00–06:00 on completed nights. Associated with a lower estimate for this meal.

lowers

-0.22

OBSERVED

Glucose and meals

Native Dexcom readings around the meal. The twin sees only what was recorded before its moment; reveal the rest to compare prediction with reality.

Model: 75% chance > 180

peak 204

180 mg/dL · outcome threshold

twin's moment

05:00 06:00 07:00 08:00 09:00 10:00 11:00

Point at or tab to a meal ▲ for its details and 2-hour response. Tap or click the chart to hold a reading; arrow keys step through readings.

Measured (native Dexcom, 5-min)

Measured later, hidden from the twin

Prediction window, 120 min

70–180 mg/dL

Meal

10:27

120 of 120 min after the twin's moment

MODEL ESTIMATE Predicted at 08:27

75% threshold 37%

MEASURED Highest reading in 2 h

204 mg/dL at 09:19 · first above 180 at 09:09

Outcome (labels.v1)

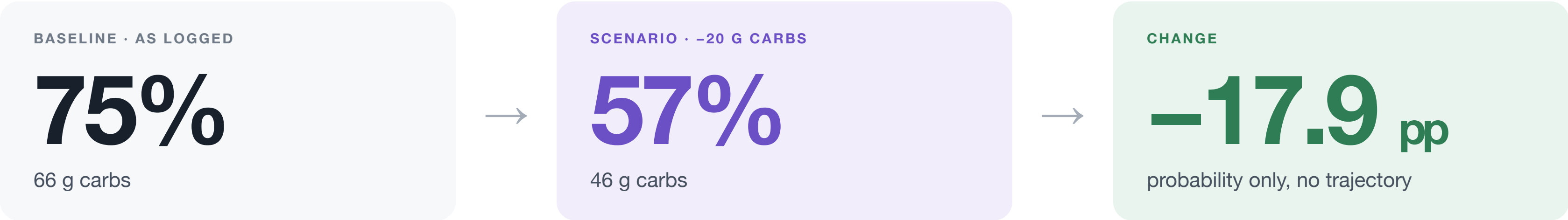
Above 180 frozen label, 1-minute series

Exact TreeSHAP on the served model: association, not cause · one example meal; aggregate performance is on slide 11

14

— WHAT-IF

Change the meal. See the **estimate** move.



Threshold

stays above the 37% alert threshold, closer to it (20.0 pp above, was 37.9)

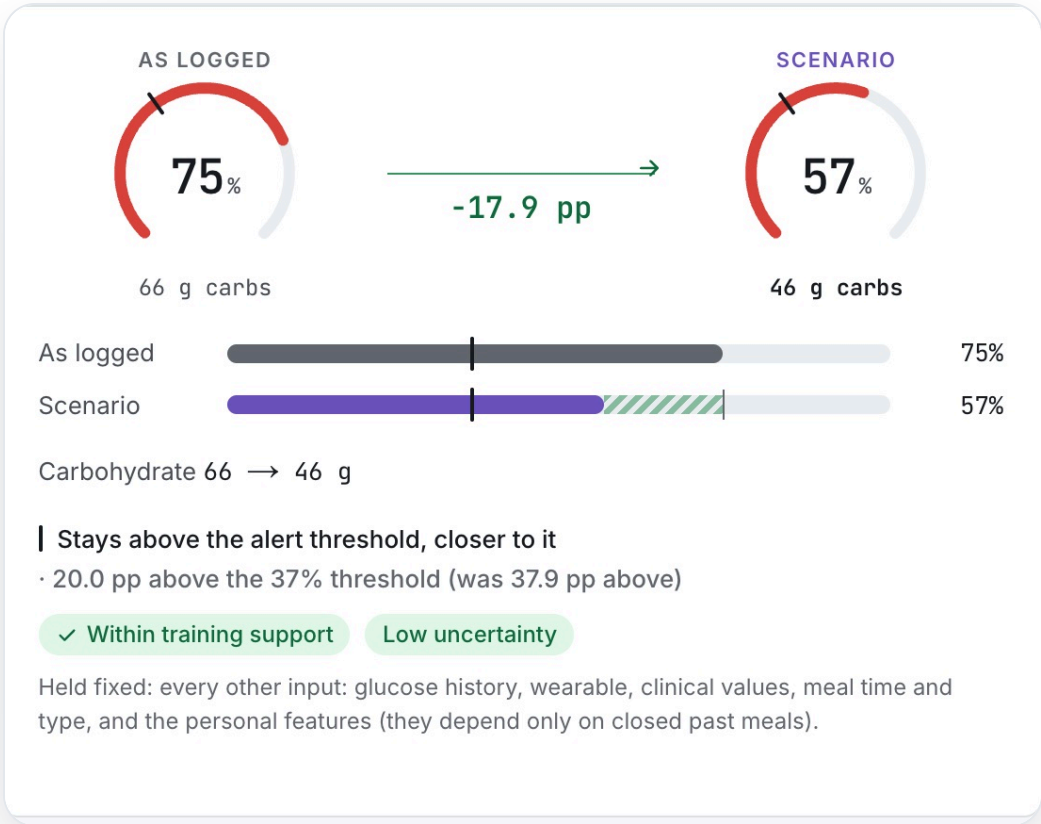
Uncertainty

low · within training support (each input inside its 1st–99th percentile)

Bounded

carbs, fiber, protein, fat, pre-meal activity only; outside support: “no estimate”

- Model-estimated
- Non-causal
- Not medical advice
- No medication or insulin levers



As **as_of** moves, the Twin evolves

t0	t1	t2	t3
Wed 11 Mar · 12:09 lunch	Wed 11 Mar · 19:38 dinner	Thu 12 Mar · 14:25 lunch	Thu 12 Mar · 20:04 dinner
COLD_START	WARMING	WARMING	PERSONALIZED
closed meals learned0	closed meals learned1	closed meals learned3	closed meals learned4
personal positive rate0.43 (= population)	personal positive rate0.63	personal positive rate0.78	personal positive rate0.65
estimate for this meal19%	estimate for this meal61%	estimate for this meal46%	estimate for this meal42%

Historical replay ⓘ

Wed 11 Mar 12:39 step 2 of 49

▶ Resume

Step >

every 30 min ▼

× Exit

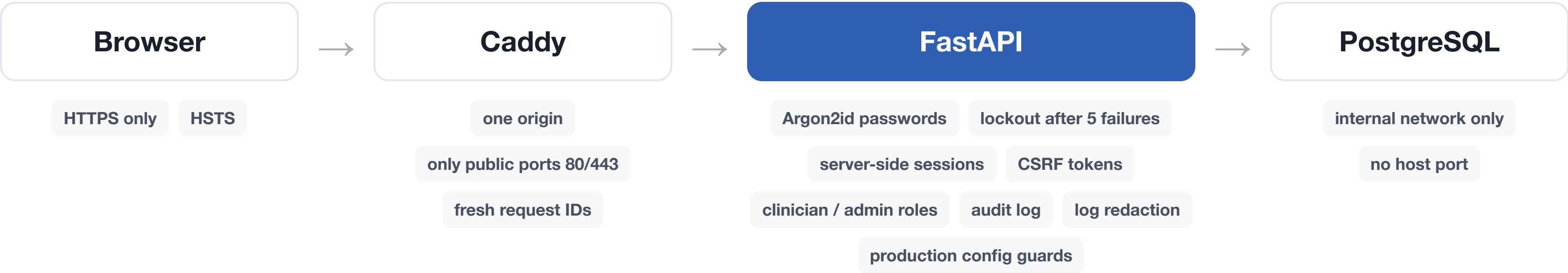
Paused. Explore the twin at this moment, then resume.

Play, pause or step: at each step the server rebuilds the Twin at the new moment, using only what was observable then. Each estimate is for the meal at that moment.

HISTORICAL REPLAY

NOT LIVE MONITORING

Clinical-grade habits in a research prototype



Non-root containers
API and web images run as unprivileged users

Google sign-in, provisioned only
no account is ever created from Google

No secrets or patient data in git
no raw data, processed tables or model bundles

A small, defensible stack, grouped by layer

DATA / ML Python 3.11 · pandas · PyArrow · scikit-learn · XGBoost · SHAP	TWIN CORE twin_core: pure Python, no web or database code	BACKEND FastAPI · Pydantic v2 · SQLAlchemy 2 · Alembic
DATABASE PostgreSQL 16	FRONTEND Next.js 15 · React 19 · TypeScript · TanStack Query · hand-written SVG charts	DELIVERY Docker (non-root) · Compose · Caddy · GitHub Actions · uv lock file

370

backend tests, incl. PostgreSQL migrations

105

frontend unit tests

51

Playwright E2E: axe accessibility, reduced motion, mobile

CI on every pull request: lint, types, tests (fails if any test is skipped), production images built for linux/amd64. No LLMs, no external AI APIs.

A complete, reproducible prototype, from open data to a clinician's screen

Real data

CGMacros v1.0.0, audited and cleaned with versioned rules

Leakage-safe ML

all leakage checks passed · PR-AUC 0.808 vs 0.701 baseline on unseen participants, calibration slope 1.00

Digital Twin

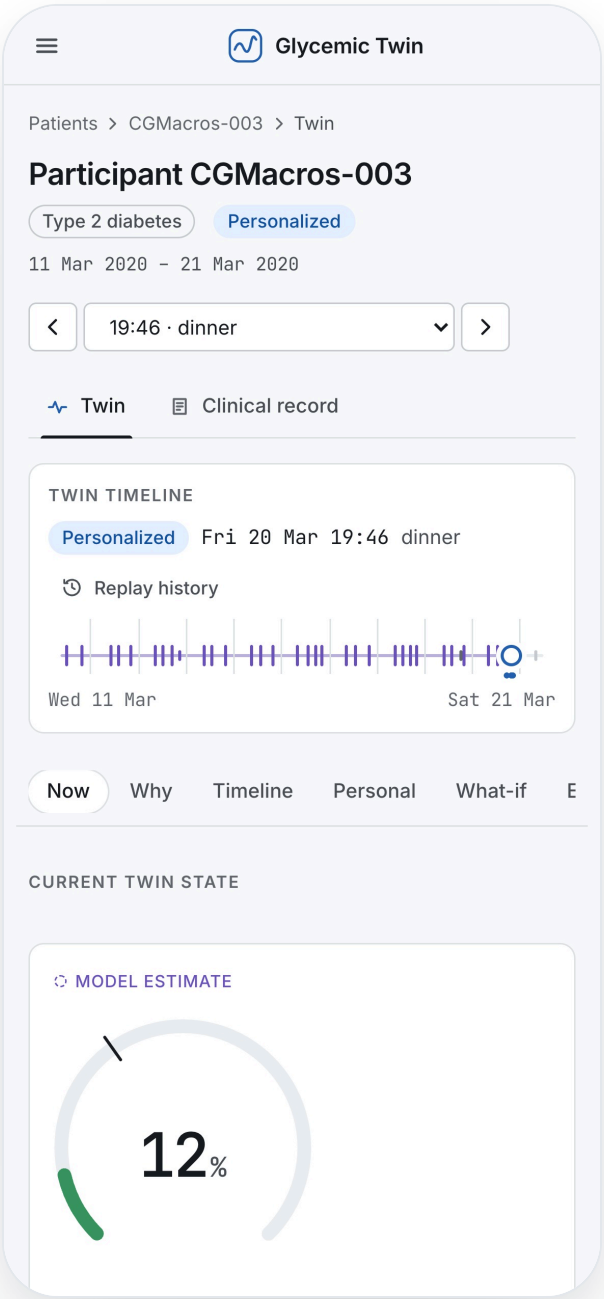
time boundary, lifecycle, provenance, what-if, historical replay

Interactive clinician tool

accessible, responsive, secure

Reproducible engineering

one-command Docker stack · green CI · MIT-licensed code



— LIMITATIONS

What this prototype does **not** claim

Small cohort

about 30 target-group participants over ~10 days: wide confidence intervals

Association $\neq$ causation

what-if and SHAP show what the model relies on, not what would happen

Limited what-if levers

current-meal macros and pre-meal activity only; no post-meal activity, medication or insulin

One evaluation regime

new participants only; temporal validation is future work

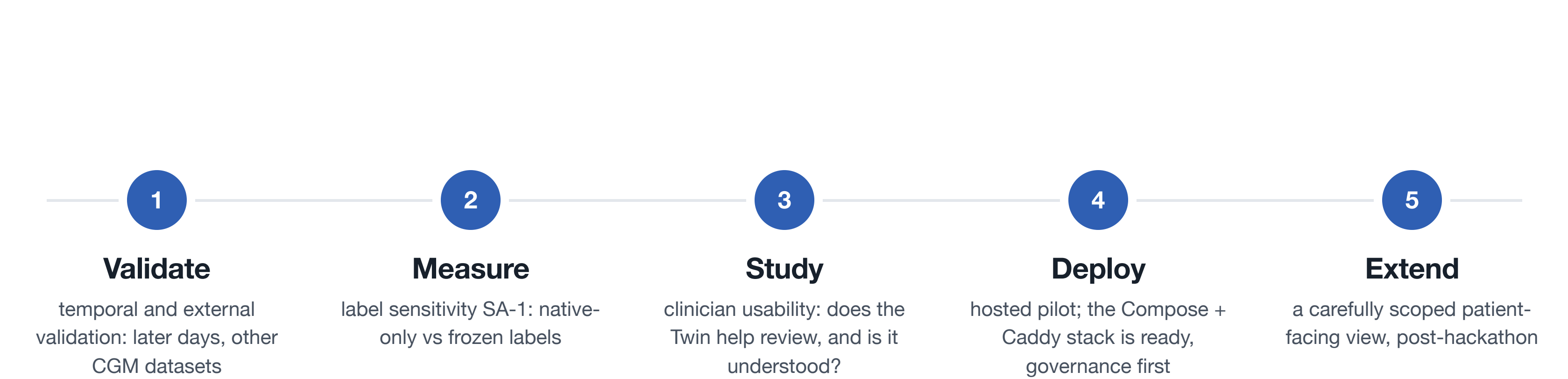
Label basis

frozen 1-minute series; interpolation effect unmeasured (SA-1)

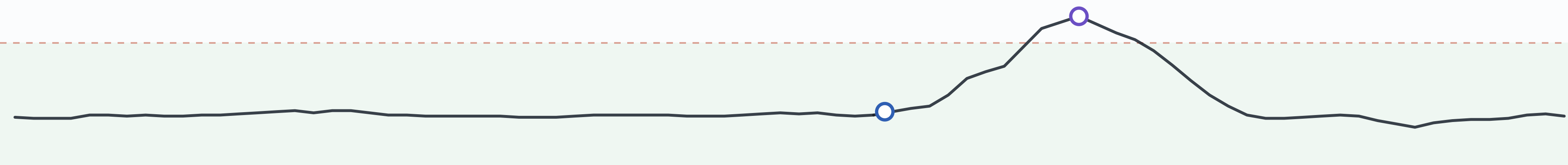
Not clinical

not a medical device, not clinically validated, not publicly deployed, not live monitoring

Next steps, evidence first



Now let me show you the **Twin**.



CODE
github.com/mitanshm18/glycemic-twin

LICENSE
Code MIT · Data CC BY-NC-SA 4.0

TEAM
IdeaForge · IIIT Allahabad